Mixed-degree cancellation: field and word proofs

The source-derived proofs exclude coincident-y mismatch from valid mixed-degree inputs and preserve canonical normalized validity on cancellation. Output identities and validity are conclusions, not premises.

Field construction. The literal guard satisfies $K_{raw} = Z^4 Z_b^4 W u(x)$. With nonzero weights, K=0 gives $u(x) = 0$ and

$$u(U) = (U - x)(U - (-q - x)).$$

Both extracted supports are valid by divisibility. Combining the original point validity with the extracted first support yields a constant divisible by a degree-one polynomial; that constant vanishes, giving $y^2 = v(x)^2$. Exact ordinate comparisons are $L = Dy$, $R = Dv(x)$, where $D = Z^5 Z_b^5 W W_b \neq 0$. Consequently the later mismatch is excluded after the earlier cancellation test.

The source p4 normalizes to $x_4 = -q - x, y_4 = v(x_4)$ and has nonzero weights ZZ_b, W_b. No division by y or $x - x_4$ is needed, so repeated-root and zero-ordinate cancellation are retained.

UInt64 composition. Exact source parentheses are modeled for the prelude and p4. Existing expression composition gives canonical words and field decoding under prime $5 \leq p < 2^{31}$ and10canonical inputs. Equality/negation reflection proves guard correspondence. The final theorem starts with both normalized validity predicates and four nonzero word weights; it excludes mismatch and proves canonical valid degree-one output with nonzero word weights on cancellation.

Controls and receipts. Full nonunit rational cancellation uses a=(9,-486,3,2), b=(-4,0,24,0,2,3), f=X^2, with normalized point(1,-1), supportX(X-1), ordinateX. A full F7 word instantiation covers repeated-root/zero-ordinate cancellation with nonunit weights and f=0. Invalid ordinate, p5, zero weights and nonzero literal payload controls are retained. These are polynomial fixtures, not smooth-curve claims.

Final field/word modules, controls and separate readback exit0. Axioms are propext, Classical.choice and Quot.sound. Failed noncomputability/tactic/simplification/syntax/control attempts are preserved with source snapshots. Compiles used pinned dependencies and120-second bounds.

Boundary. The pure bundle may compute unused p4 values; no cost or side-effect correspondence is asserted. Common and positiveCoincidence denote prelude tails only; generic, tripling and secondary outputs remain separate. Field.n=1, tables=None and counter headroom are external. No compiler/counter, class-sum or whole-dispatcher claim. Independent review pending; no scientific trials or costs. Sources: exact gate828–878, prior plan and Ozbay Gurler–Hisil sections2,3.3,4.2, DOI10.1007/s13389-024-00351-7.